

Evaluation of Algorithms for Fractal Dimension Estimation in Medical Images: Complexity and Performance Analysis

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Resumo

Resumo do artigo em português

Palavras-chave: Palavras chave em português

Resumo

Abstract

Resumo do artigo em inglês

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1 Introduction

Fractal geometry, formalised by Benoit B. Mandelbrot in his landmark monograph *The Fractal Geometry of Nature* (MANDELBROT, 1982), provided a mathematical framework for describing the roughness, branching, and statistical self-similarity pervasive in natural phenomena that classical Euclidean geometry fails to capture. The central quantitative descriptor of this framework is the fractal dimension D , estimated in practice as the negative slope of a linear regression over a log-log plot of a covering measure against the measurement scale ε . Shortly after Mandelbrot’s unification, Pentland (PENTLAND, 1984) demonstrated that fractal dimension could be computed directly from digital imagery to discriminate surface textures in natural scenes — a result that bridged fractal geometry and image analysis, and opened the way for its systematic application to medical imaging tasks.

The adoption of fractal analysis in medical imaging occurs since the 1990s, alongside the proliferation of high-resolution digital modalities — multi-detector computed tomography, 3 T magnetic resonance imaging, and digital mammography. Applications spanning pulmonary emphysema (MISHIMA *et al.*, 1999), retinal vascular morphology (JELINEK; FERNANDEZ, 2000), brain cortical folding (FREE *et al.*, 1996), and tumour margin irregularity (RANGAYYAN; NGUYEN, 2007) demonstrated that D encodes structural properties that classical first-order statistical descriptors — variance, entropy, autocorrelation — could not adequately capture. From the 2010s onward, fractal descriptors were incorporated into hybrid feature engineering pipelines alongside learned representations (LOPES; BETROUNI, 2011; KING *et al.*, 2010), and more recently, fractal dimension has been explored as a feature describer to enhance the interpretability of deep learning models (ROBERTO *et al.*, 2021) and as

a data augmentation strategy with GANs (BORGUE *et al.*, 2025). Contemporary clinical datasets — commonly comprising images at resolutions exceeding 1024×1024 pixels — elevated the algorithmic efficiency of dimension estimation from a secondary concern to a primary design constraint. The central question shifted from whether fractal dimension is discriminative to at what computational cost it can be reliably extracted, and which algorithm offers the optimal balance of precision, robustness, and efficiency under realistic operating conditions.

Addressing this question requires a rigorous algorithmic engineering perspective. The four methods selected for analysis span the principal design axes of the problem: binary versus grayscale input, and non-overlapping versus dense-overlap covering strategies. The *Classical Box-Counting* (BC) (MANDELBROT, 1982) partitions the binary image into a grid of non-overlapping boxes of decreasing side and counts how many contain foreground pixels, yielding a coarse but efficient estimate of D . The *Gliding Box* (GB) (ALLAIN; CLOITRE, 1991) extends this idea by sliding a window across every pixel position, producing a richer mass-frequency distribution at the cost of processing a far larger number of box placements. The *Differential Box-Counting* (DBC) (SARKAR; CHAUDHURI, 1994) lifts the problem to grayscale images by treating the intensity surface as a three-dimensional relief and counting the column of boxes required to span the local intensity range within each cell, thereby retaining gradient information that binary methods discard. The *Blanket Method* (BM) (PELEG *et al.*, 1984) estimates the fractal dimension of the intensity surface by measuring how its apparent area grows as a morphological cover is progressively inflated and deflated around it at each scale. Contrasting these four methods also surfaces a fundamental trade-off inherent to the pre-processing stage: binarisation lowers the representational

cost of the input, but discards intensity gradients that carry clinically significant discriminatory information — a loss that the grayscale methods avoid at the price of higher computational demand.

This work addresses the following research objectives:

- **Algorithmic analysis:** for each of the four methods, in both naive and structurally optimised formulations, derive formal asymptotic bounds for time and auxiliary space complexity and validate them against empirical measurements on the UCSB medical image dataset, contrasting theoretical scaling predictions with observed wall-clock behaviour.
- **Pre-processing and implementation cost:** quantify the overhead imposed by mandatory pre-processing stages and characterise the data-structure choices that drive the efficiency gap between naive and optimised implementations.
- **Effectiveness and trade-off synthesis:** assess the discriminatory power of each algorithm by measuring the statistical separation it induces between pathological and healthy tissue classes in terms of the estimated fractal dimension, and synthesise all time, memory, and effectiveness evidence into a structured trade-off comparison across the binary-versus-grayscale and naive-versus-optimised design axes.

The remainder of this article is structured as follows: Section 2 establishes the mathematical foundations of fractal analysis and formally defines each algorithm; Section 3 presents pseudocode and derives asymptotic complexity bounds; Section 4 describes the experimental methodology, dataset, and evaluation metrics; Section 5 reports and discusses empirical results; Section 6 concludes with recommendations and directions for future work.

2 Theoretical Foundation

- **Item1** Descrição.
- **Item2** Descrição.

$$C(i) = i + 3 - \frac{1}{3} \sum_k^{10} 2k \quad (1)$$

1. Item1;
2. Item2;
3. Item3;

3 Algorithms Analysis

4 Methodology

5 Results and Discussion

Resultados da pesquisa

Tabela 1: Legenda

col1	bol2	bol3	col4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
dado1	dado2	dado3	dado4
resultado1	resultado1	resultado1	resultado1

nota de rodapé

6 Conclusion

Retomada da proposta, resultados e possíveis melhorias/trabalhos futuros.

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